

ShareGo - FYP Evaluation Cheatsheet

1. Project Overview

ShareGo is a peer-to-peer mobile marketplace for Pakistan with two core features: Feature A (Cross-Border Delivery) lets buyers request products from abroad while verified travelers carry them using spare luggage capacity, protected by simulated escrow with OTP-verified handover. Feature B (Local Marketplace) is an OLX-style buy/sell platform with offer bargaining and meetup scheduling. Built with Flutter (Android) + FastAPI (Python) + SQLite, with a multi-LLM AI assistant and full admin dashboard.

2. Tech Stack

Layer	Technology	Purpose
Mobile	Flutter 3.3+ / Dart	Android app (Material Design 3)
State Mgmt	Riverpod 3.x	Reactive state management
Navigation	Go Router 17.x	Declarative routing
HTTP Client	Dio 5.x	API calls with retry logic
Backend	FastAPI 0.110 / Python	REST API + admin panel
ORM	SQLModel + SQLAlchemy	Database models & queries
Database	SQLite (WAL mode)	Persistent storage
Migrations	Alembic	Schema version control (10 revisions)
Admin Panel	Jinja2 Templates	Server-rendered HTML dashboard
Auth	JWT (HS256) + bcrypt OTP	Token-based + one-time passwords
AI	Gemini 2.5 + NVIDIA + OpenAI	Multi-provider LLM with fallback
Flight Tracking	AviationStack API	Real-time flight status
Rate Limiting	SlowAPI	DDoS protection
Theme	Material 3 (Dark + Light)	Dark/light mode toggle
Refresh Rate	flutter_displaymode	120Hz support
Google Auth	Google Sign-In (OAuth 2.0)	Social login

3. Architecture

Three-layer architecture:

- Client: Flutter Mobile App (Riverpod + Go Router + Dio HTTP)
- Backend: FastAPI REST API + Jinja2 Admin Panel + SQLite (WAL) + Alembic
- External: Google Gemini, NVIDIA NIM, OpenAI, AviationStack, Google OAuth, SMTP
- Security: JWT Auth, OTP (bcrypt), SlowAPI Rate Limiting, CORS, Audit Logging

4. Feature A - Cross-Border Delivery

Buyer Journey:

- 1. Post Request - Item name, specs, target price, weight, time window
- 2. Browse Travelers - Filter by route, date, capacity
- 3. Select Traveler - View profile, reviews, flight details
- 4. Create Booking - Enter item details
- 5. Escrow Hold - Simulated payment deducted from wallet
- 6. Wait for Acceptance - Traveler accepts or declines
- 7. Track Status - Traveler posts updates (picked up, airport, in transit, arrived)
- 8. Delivery OTP - Enter OTP to confirm delivery + GPS + photos
- 9. Escrow Released - Admin releases funds to traveler
- 10. Leave Review - Rate traveler (1-5 stars)

Traveler Journey:

- 1. KYC Verification - Upload CNIC/Passport + Selfie (admin approves)
- 2. Post Trip - Flight details, route, date, capacity, fee/kg
- 3. Admin Approval - Trip reviewed (pending_review -> approved)
- 4. Accept Booking - Review request, accept or decline
- 5. Purchase Item - Buy abroad, upload receipt
- 6. Post Status Updates - Picked up, at airport, in transit, arrived
- 7. Pickup OTP - Verify with OTP + GPS + seal ID
- 8. Deliver Item - Meet buyer, verify with delivery OTP
- 9. Get Paid - Admin releases escrow to wallet
- 10. Leave Review - Rate buyer

Booking State Machine:

REQUESTED -> HOLD_PLACED -> ACCEPTED -> PICKUP_OK -> DELIVERY_OK -> RELEASED

Terminal states: CANCELLED, EXPIRED

5. Feature B - Local Marketplace

Seller Journey:

- Create Listing - Title, photos, ask price, category, location (GPS)
- Receive Offers - Buyers send price offers with messages
- Bargain - Counter-offer, accept, or decline
- Schedule Meetup - Set place and time
- Meet and Sell - Confirm meetup with OTP
- Mark Sold - Listing removed from feed
- Leave Review - Rate the buyer

Offer State Machine:

SENT -> COUNTERED -> ACCEPTED / DECLINED / WITHDRAWN

6. Database Tables (20 Tables)

#	Table	Purpose
1	User	Accounts: email, password, KYC status, roles, rating
2	OTPEntry	OTP verification (bcrypt-hashed, single-use)
3	OTPThrottle	Rate limiting OTP per email + IP
4	Trip	Traveler trips: flight, capacity, status
5	KYCProfile	Identity docs (CNIC/Passport + Selfie + Passport)
6	RequestItem	Buyer product requests with specs
7	Booking	Feature A bookings with state machine + OTP hashes
8	BookingUpdate	Traveler status updates timeline
9	EscrowTx	Simulated escrow (hold/released/refunded)
10	WalletEntry	Virtual wallet ledger (credits/debits)
11	HandoverEvent	Pickup/delivery verification (OTP+GPS+photos)
12	MarketListing	Feature B listings with photos, price, location
13	MarketOffer	Offer bargaining (send/counter/accept/decline)
14	MarketMeetup	Meetup scheduling with OTP confirmation
15	Message	In-app chat linked to booking/listing
16	Review	Ratings (1-5 stars) for bookings and listings
17	Dispute	Dispute tracking for problematic bookings
18	DisputeEvidence	Evidence files for disputes
19	FeatureFlag	Feature toggles (enable/disable by cohort)
20	AuditLog	Audit trail: actor, action, before/after snapshots

7. API Endpoints (50+)

- Auth (6): register, register-password, verify-otp, login, google, forgot
- Trips (5): CRUD + flight-status
- Requests (4): CRUD
- Bookings (10): CRUD + accept/decline/cancel + pickup/delivery + updates
- Marketplace (10): listings CRUD + offers + counter/accept/decline + meetup + sold
- KYC (2): submit + status
- Reviews (2): submit + list
- Messages (2): send + list
- AI (1): chat
- Media (1): upload
- Admin (15+): KYC, bookings, escrow, trips, users, marketplace, wallets, audit, flags

8. Security Measures

- JWT authentication (HS256, 60-minute expiry)
- Password hashing: bcrypt (12 rounds)
- OTP: bcrypt-hashed, single-use, 300s TTL, 3 max attempts
- Rate limiting: 3 OTP/min, 5 login/min, 10 Google/min, 30 AI/hour
- OTP throttling per email + IP address
- CORS middleware with configurable origins
- TrustedHostMiddleware for host validation
- Admin cookie auth with session timeout
- Owner/role-based access guards on all endpoints
- Audit logging for all mutations (actor, before/after)
- Request ID tracking on every HTTP request
- Pydantic schema validation on all inputs
- File upload: MIME type + extension check (jpg/png only)
- Waybill format validation (SG-XXXX-XXXXXX)

9. Admin Panel Features

- Dashboard with KPI overview
- KYC Queue - Approve/reject with inline image thumbnails
- Bookings - Timeline, detail page, escrow release/refund
- Trips - Approve/reject with reason
- Users - Make admin, remove admin, delete user
- Marketplace - Flag/unflag listings
- Wallets - View balances, admin top-up
- Audit Logs - Filter by action/admin/entity; before/after snapshots
- Feature Flags - Enable/disable by cohort

10. AI Assistant

- Multi-LLM: NVIDIA gpt-oss-120b -> OpenAI gpt-4o-mini -> Gemini 2.5 Flash
- Travel Safety - War zones, safe countries, visa info
- Courier Guidance - Items per country, customs rules
- Item Legality - Risk levels (LOW/MEDIUM/HIGH/PROHIBITED)
- Shipping Calculator - Fee estimates by weight
- FAQ - Escrow, KYC, marketplace rules
- DB Tools (read-only) - Bookings, trips, listings, offers
- Rate Limited - 30 requests/hour per user

11. SDG Alignment

- SDG 8 (Decent Work): Income for travelers via spare luggage
- SDG 12 (Responsible Consumption): Sharing economy optimizes travel capacity

12. Key Terminology

Term	Meaning
Feature A	Cross-border personal shopping with escrow simulation
Feature B	Local OLX-style marketplace with bargaining
Escrow	Simulated payment hold (virtual wallet, not real money)
KYC	Know Your Customer - identity verification
Waybill	Booking reference (SG-XXXX-XXXXXX)
Seal ID	Physical tamper-proof seal verified at pickup/delivery
OTP	One-Time Password for auth AND handover verification
Handover	Physical exchange verified with OTP + GPS + photos
WAL Mode	SQLite concurrent reads during writes
Riverpod	Flutter reactive state management
JWT	JSON Web Token - stateless authentication
Alembic	Python database migration tool
Crowdshipping	Peer-to-peer delivery using travelers

13. Evaluation Q&A

Q: What problem does ShareGo solve?

A: Pakistan lacks a P2P platform connecting travelers with buyers needing overseas products. Traditional shipping is expensive. ShareGo lets travelers earn from spare luggage while buyers get cheaper products.

Q: Why not Grabr or PiggyBee?

A: They don't target Pakistan, don't support PKR, lack CNIC-based KYC, and have no local marketplace.

Q: How do you handle trust?

A: KYC verification, escrow hold, OTP-verified handover with GPS+photos, seal ID matching, ratings/reviews, dispute system, admin oversight.

Q: Is the escrow real money?

A: No - simulated with virtual wallet (WalletEntry table). Production would use Stripe/JazzCash.

Q: How does the AI work?

A: Multi-provider: NVIDIA gpt-oss-120b (primary) -> OpenAI -> Gemini (fallback). Strict system prompt, read-only DB tools, rate limited 30/hour.

Q: Why SQLite not PostgreSQL?

A: SQLite with WAL mode is sufficient for FYP. Zero config, no server. Alembic migrations make switching to PostgreSQL trivial.

Q: How do you prevent illegal items?

A: Policy flags, AI customs guidance, admin moderation, KYC identity verification. Users responsible for customs compliance.

Q: Testing strategy?

A: Integration tests: auth OTP flow, booking lifecycle, marketplace flow, ownership guards (403), rate limiting (429), invalid transitions (400).

Q: Production improvements?

A: Real payments (Stripe/JazzCash), PostgreSQL, WebSocket chat, push notifications, cloud deployment, automated KYC (OCR+face), E2E encryption, ML matching.

Q: Which SDGs?

A: SDG 8 (Decent Work): Income for travelers. SDG 12 (Responsible Consumption): Sharing economy.

Q: How does dark theme work?

A: Material 3 ColorScheme.fromSeed() with Brightness.dark. All screens use Theme.of(context). Persisted in SharedPreferences.

Q: How are booking notifications sent?

A: Traveler posts status update -> server auto-creates Message to buyer. Buyer sees it in Messages tab.

14. Project Statistics

Metric	Count
Dart files	40+
Python files	30+
API endpoints	50+
Database tables	20
Alembic migrations	10
Admin templates	11
Mobile screens	30+
Server code	~5,000 lines
Mobile code	~8,000 lines